1. **Write a program to implement removal of recursion for – Finding maximum from array.**

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter number: ";
    cin >> n;

    int arr[n];

    cout << "Enter elements: ";
    for (int i = 0; i < n; i++)
        cin >> arr[i];

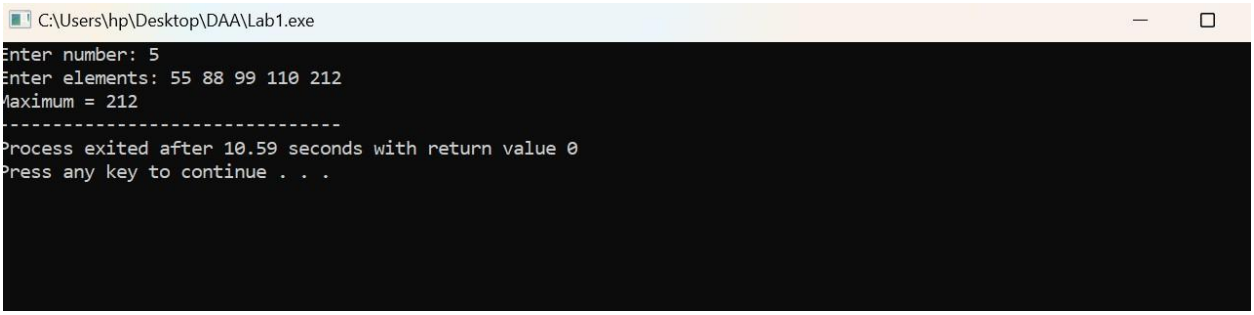
    int max = arr[0];

    for (int i = 1; i < n; i++)
        if (arr[i] > max)
            max = arr[i];

    cout << "Maximum = " << max;

    return 0;
}
```

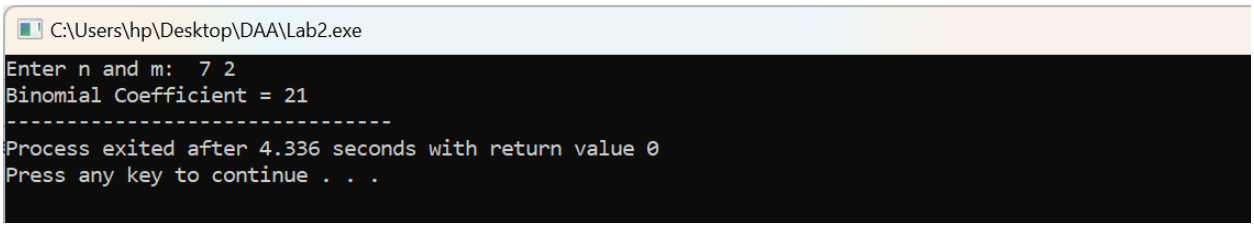
Output:



2. Write a program to implement removal of recursion for – Binomial Coefficient B(n,m)= B(n-1, m-1)+B(n-1,m), B(n,n)=B(n,0)=1

```
#include  <iostream>
using namespace std;
int main() {
    int n, m;
    cout << "Enter n and m: ";
    cin >> n >> m;
if (m < 0 || m > n) {
        cout << "Invalid input";
        return 0;}
int C[n + 1][m + 1];
for (int i = 0; i <= n; i++) {
        for (int j = 0; j <= m && j <= i; j++) {
           if (j == 0 || j == i)
              C[i][j] = 1;
           else
              C[i][j] = C[i - 1][j - 1] + C[i - 1][j];
        } }
 cout << "Binomial Coefficient = " << C[n][m];
 return 0; }
```

Output:



3. Write a program to implement removal of recursion for – Searching element from array.

```
#include  <iostream>

using namespace std;

int main() {

    int n, key;
    cout << "Enter number of elements: ";

    cin >> n;

 int arr[n];
cout << "Enter array elements: ";

    for (int i = 0; i < n; i++)

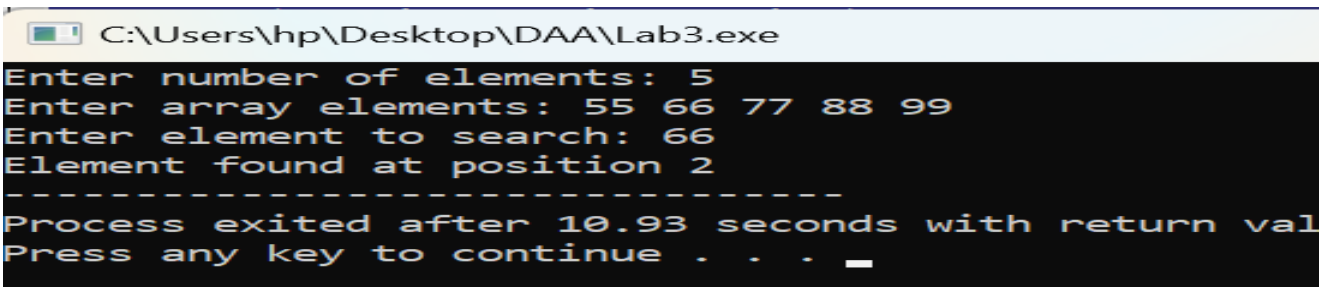
        cin >> arr[i];
 cout << "Enter element to search: ";

    cin >> key;

 int pos = -1;
 for (int i = 0; i < n; i++) {
        if (arr[i] == key) {
            pos = i;
            break;
        }}
if (pos != -1)

        cout << "Element found at position " << pos + 1;
    else
        cout << "Element not found";
return 0;
}
```

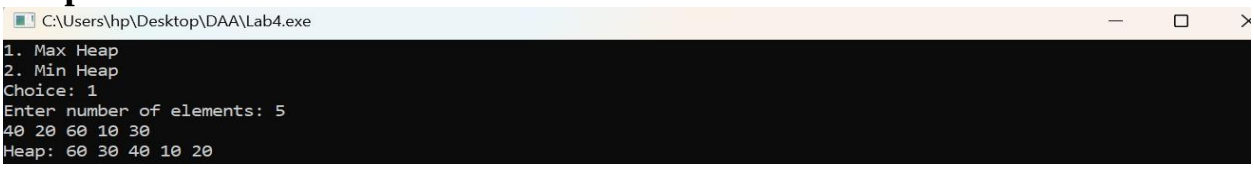
Output:



4. Write a program for creating Max/Min. heap using INSERT.

```
#include <iostream>
using namespace std;
void insert(int h[], int &n, int x, bool maxHeap) {
    h[++n] = x;
    int i = n;
 while (i > 1 && (maxHeap ? h[i] > h[i/2] : h[i] < h[i/2])) {
        swap(h[i], h[i/2]);
        i /= 2;
  }}
int main() {
    int h[100], n = 0, x, choice, size;
 cout << "1. Max Heap\n2. Min Heap\nChoice: ";
  cin >> choice;
cout << "Enter number of elements: ";
  cin >> size;
 for (int i = 0; i < size; i++) {
        cin >> x;
        insert(h, n, x, choice == 1);}
cout << "Heap: ";
    for (int i = 1; i <= n; i++)
        cout << h[i] << " ";
 return 0;  }
```

Output:

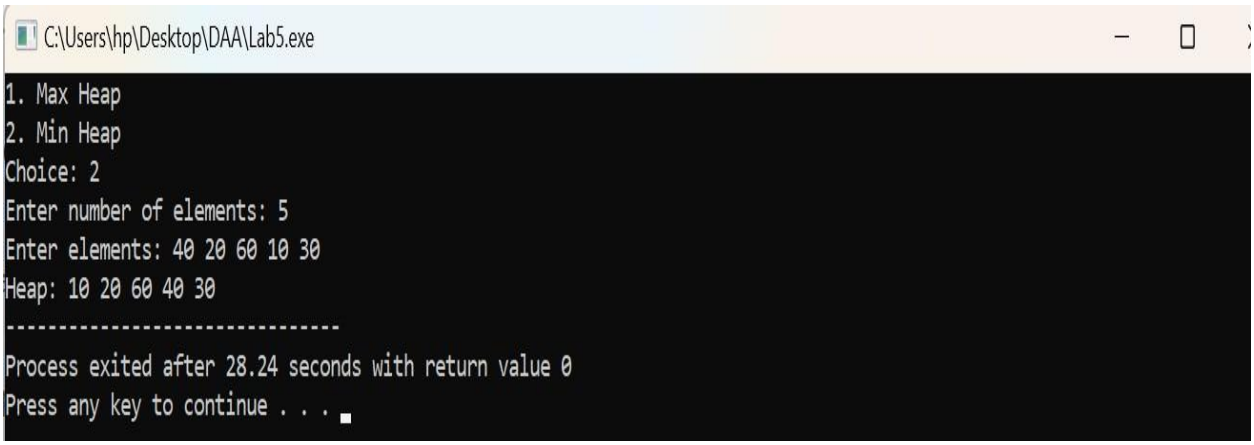


5. Write a program for creating Max/Min. heap using ADJUST/HEAPIFY.

```
#include <iostream>
using namespace std;
void heapify(int h[], int n, int i, int max) {
    int x = i, l = 2*i, r = 2*i+1;
 if (l <= n && (max ? h[l] > h[x] : h[l] < h[x]))
        x = l;
if (r <= n && (max ? h[r] > h[x] : h[r] < h[x]))
        x = r;
if (x != i) {
        swap(h[i], h[x]);
        heapify(h, n, x, max);
    }}
int main() {
    int h[100], n, choice;
  cout << "1. Max Heap\n2. Min Heap\nChoice: ";
    cin >> choice;
cout << "Enter number of elements: ";
    cin >> n;
 cout << "Enter elements: ";
    for (int i = 1; i <= n; i++)
        cin >> h[i];
for (int i = n/2; i >= 1; i--)
        heapify(h, n, i, choice == 1);
  cout << "Heap: ";
    for (int i = 1; i <= n; i++)
        cout << h[i] << " ";
```

```
return 0;
}
```

Output:

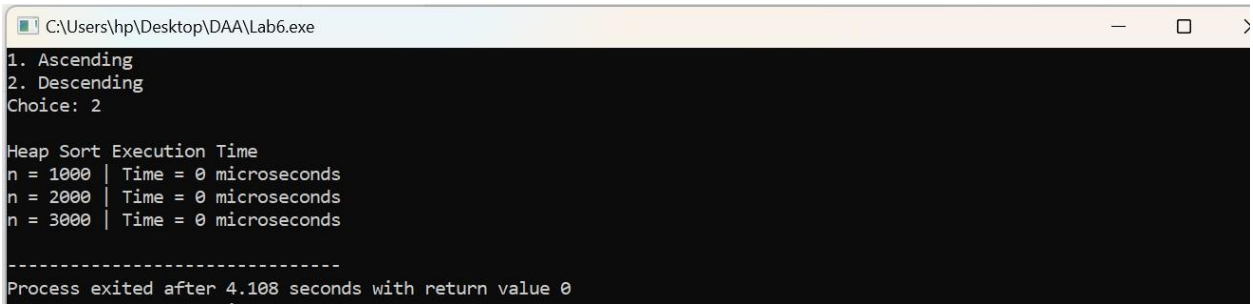


6. Write a program for sorting given array in ascending/descending order with n=1000,2000,3000. Find exact time of execution using Heap Sort.

```
#include <iostream>
#include <chrono>
using namespace std;
using namespace chrono;
void heapify(int a[], int n, int i, bool asc) {
   int x = i, l = 2*i+1, r = 2*i+2;
 if (l < n && (asc ? a[l] > a[x] : a[l] < a[x]))
       x = l;
 if (r < n && (asc ? a[r] > a[x] : a[r] < a[x]))
       x = r;
 if (x != i) {
       swap(a[i], a[x]);
       heapify(a, n, x, asc);
   }}
void heapSort(int a[], int n, bool asc) {
   for (int i = n/2-1; i >= 0; i--)
       heapify(a, n, i, asc);
 for (int i = n-1; i > 0; i--) {
       swap(a[0], a[i]);
       heapify(a, i, 0, asc);
   }}
void test(int n, bool asc) {
   int *a = new int[n];
  for (int i = 0; i < n; i++)
 a[i] = n - i;
```

```
auto start = high_resolution_clock::now();
 heapSort(a, n, asc);
 auto end = high_resolution_clock::now();
cout << "n = " << n << " | Time = "
     << duration_cast<microseconds>(end-start).count()
       << " microseconds\n";
delete[] a;
}
int main() {
    int choice;
    cout << "1. Ascending\n2. Descending\nChoice: ";
    cin >> choice;
   bool asc = (choice == 1);
 cout << "\nHeap Sort Execution Time\n";
    test(1000, asc);
    test(2000, asc);
    test(3000, asc);
 return 0;
}
```

Output:

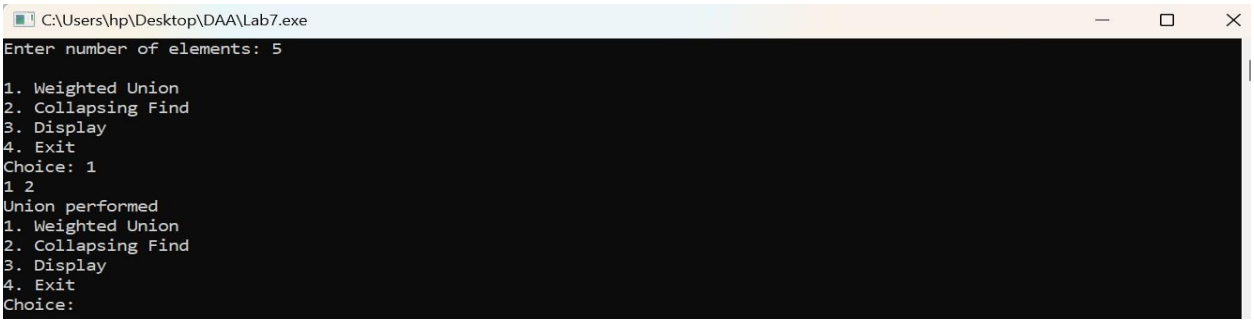


7. Write a program to implement Weighted UNION and Collapsing FIND operations.

```
#include <iostream>
using namespace std;
int parent[100], sizeSet[100];
int find(int x) {
    if (parent[x] != x)
        parent[x] = find(parent[x]);   // Collapsing Find
    return parent[x];
}void weightedUnion(int a, int b) {
    a = find(a);
    b = find(b);
 if (a == b) return;
 if (sizeSet[a] < sizeSet[b]) {
        parent[a] = b;
        sizeSet[b] += sizeSet[a];
    } else {
        parent[b] = a;
        sizeSet[a] += sizeSet[b];
    }}int main() {
    int n, ch, a, b;
cout << "Enter number of elements: ";
    cin >> n;
 for (int i = 0; i < n; i++) {
  parent[i] = i;
        sizeSet[i] = 1;
    } do {
```

```
        cout << "\n1. Weighted Union";
        cout << "\n2. Collapsing Find";
        cout << "\n3. Display";
        cout << "\n4. Exit";
        cout << "\nChoice: ";
        cin >> ch;
if (ch == 1) {
            cin >> a >> b;
            weightedUnion(a, b);
            cout << "Union performed";
        }else if (ch == 2) {
            cin >> a;
            cout << "Root = " << find(a);
        }else if (ch == 3) {
            for (int i = 0; i < n; i++)
                cout << parent[i] << " ";
        }} while (ch != 4);
return 0;
}
```

Output:

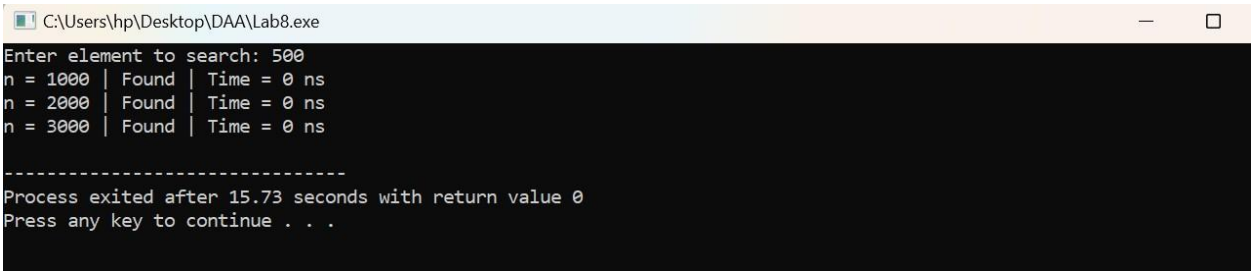


8. Write a program for searching element form given array using binary search forn=1000,2000,3000. Find exact time of execution.

```
#include <iostream>
#include <chrono>
using namespace std;
using namespace chrono;
int binarySearch(int a[], int n, int key) {
    int l = 0, h = n - 1;
 while (l <= h) {
     int m = (l + h) / 2;
 if (a[m] == key)
            return m;
        if (a[m] < key)
            l = m + 1;
        else
            h = m - 1;
    }return -1;
}void test(int n, int key) {
    int *a = new int[n];
 for (int i = 0; i < n; i++)
        a[i] = i + 1;
 auto start = high_resolution_clock::now();
    int result = binarySearch(a, n, key);
    auto end = high_resolution_clock::now();
  cout << "n = " << n
       << " | " << (result != -1 ? "Found" : "Not Found")
    << " | Time = "
```

```
        << duration_cast<nanoseconds>(end - start).count()
        << " ns\n";
 delete[] a;
}
int main() {
    int key;
    cout << "Enter element to search: ";
    cin >> key;
test(1000, key);
  test(2000, key);
  test(3000, key);
  return 0;
}
```

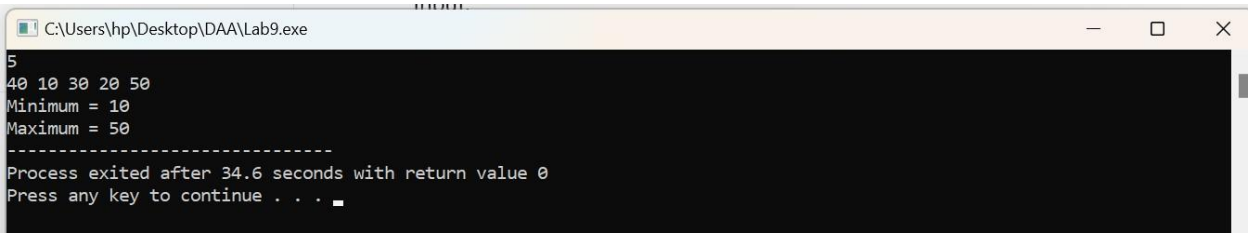
Output:



9. Write a program to find minimum and maximum from a given array using MAXMIN.

```
#include <iostream>
using namespace std;
void maxmin(int a[], int l, int h, int &mn, int &mx) {
    if (l == h)
        mn = mx = a[l];
    else {
        int m = (l + h) / 2;
        int mn1, mx1;
 maxmin(a, l, m, mn, mx);
        maxmin(a, m + 1, h, mn1, mx1);
 if (mn1 < mn) mn = mn1;
        if (mx1 > mx) mx = mx1;
    }}
int main() {
    int n;
    cin >> n;
 int a[n];
    for (int i = 0; i < n; i++)
        cin >> a[i];
int mn, mx;
    maxmin(a, 0, n - 1, mn, mx);
 cout << "Minimum = " << mn << endl;
    cout << "Maximum = " << mx;
return 0;
}
```

Output:


C:\Users\hp\Desktop\DAA\Lab9.exe
5
40 10 30 20 50
Minimum = 10
Maximum = 50

Process exited after 34.6 seconds with return value 0
Press any key to continue . . .

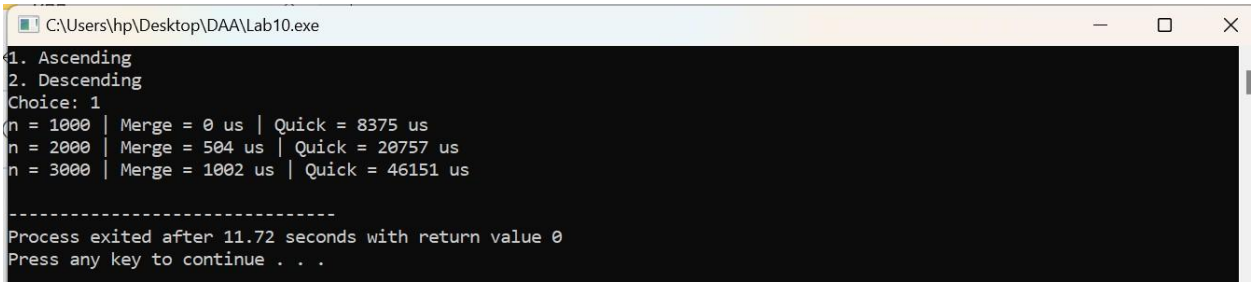
10. Write a program for sorting given array in ascending/descending order with n=1000,2000,3000 find exact time of execution using a) Merge Sort, b) Quick Sort.

```
#include <iostream>
#include <chrono>
using namespace std;
using namespace chrono;
void merge(int a[], int l, int m, int h, bool asc) {
    int i = l, j = m + 1, k = 0;
    int t[h - l + 1];
 while (i <= m && j <= h)
        t[k++] = (asc ? a[i] <= a[j] : a[i] >= a[j]) ? a[i++] : a[j++];
while (i <= m) t[k++] = a[i++];
    while (j <= h) t[k++] = a[j++];
for (i = l, k = 0; i <= h; i++, k++)
        a[i] = t[k];
}
void mergeSort(int a[], int l, int h, bool asc) {
    if (l < h) {
        int m = (l + h) / 2;
        mergeSort(a, l, m, asc);
        mergeSort(a, m + 1, h, asc);
        merge(a, l, m, h, asc);
    }}
void quickSort(int a[], int l, int h, bool asc) {
    if (l >= h) return;
  int p = a[h], i = l - 1;
```

```
for (int j = l; j < h; j++)
        if (asc ? a[j] <= p : a[j] >= p)
            swap(a[++i], a[j]);
 swap(a[i + 1], a[h]);
    int x = i + 1;
 quickSort(a, l, x - 1, asc);
    quickSort(a, x + 1, h, asc);
}
void test(int n, bool asc) {
    int *a = new int[n], *b = new int[n];
 for (int i = 0; i < n; i++)
        a[i] = b[i] = n - i;
 auto s = high_resolution_clock::now();
    mergeSort(a, 0, n - 1, asc);
    auto e = high_resolution_clock::now();
    auto mt = duration_cast<microseconds>(e - s).count();
s = high_resolution_clock::now();
    quickSort(b, 0, n - 1, asc);
    e = high_resolution_clock::now();
    auto qt = duration_cast<microseconds>(e - s).count();
 cout << "n = " << n
         << " | Merge = " << mt << " us"
         << " | Quick = " << qt << " us\n";
 delete[] a;
delete[] b;
}int main() {
    int choice;
```

```
    cout << "1. Ascending\n2. Descending\nChoice: ";
    cin >> choice;
 bool asc = choice == 1;
 test(1000, asc);
    test(2000, asc);
    test(3000, asc);
return 0;
}
```

Output:

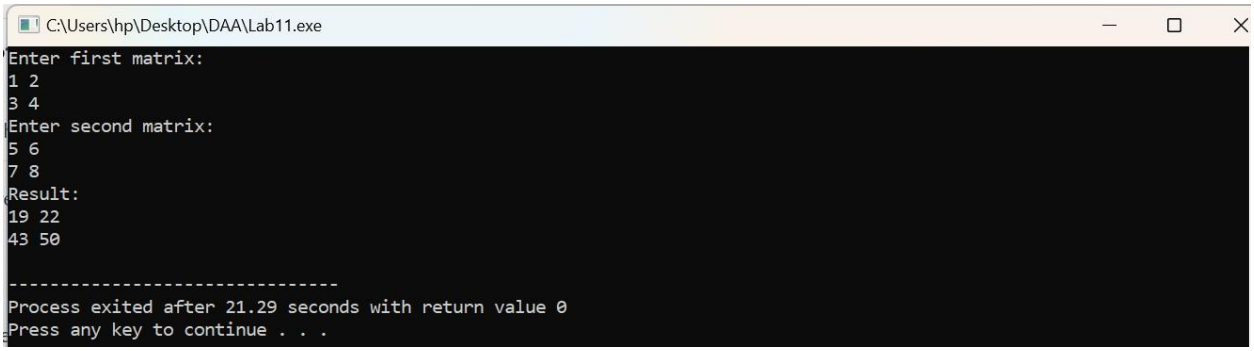


11. Write a program for matrix multiplication using Strassen's Matrix Multiplication.

```
#include  <iostream>
using namespace std;
int main() {
    int A[2][2], B[2][2], C[2][2];
    int p1, p2, p3, p4, p5, p6, p7;
 cout << "Enter first matrix:\n";
    for (int i = 0; i < 2; i++)
        for (int j = 0; j < 2; j++)
            cin >> A[i][j];
  cout << "Enter second matrix:\n";
    for (int i = 0; i < 2; i++)
        for (int j = 0; j < 2; j++)
            cin >> B[i][j];
p1 = A[0][0] * (B[0][1] - B[1][1]);
    p2 = (A[0][0] + A[0][1]) * B[1][1];
    p3 = (A[1][0] + A[1][1]) * B[0][0];
    p4 = A[1][1] * (B[1][0] - B[0][0]);
    p5 = (A[0][0] + A[1][1]) * (B[0][0] + B[1][1]);
    p6 = (A[0][1] - A[1][1]) * (B[1][0] + B[1][1]);
    p7 = (A[0][0] - A[1][0]) * (B[0][0] + B[0][1]);
C[0][0] = p5 + p4 - p2 + p6;
    C[0][1] = p1 + p2;
    C[1][0] = p3 + p4;
    C[1][1] = p5 + p1 - p3 - p7;
```

```
    cout << "Result:\n";
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++)
            cout << C[i][j] << " ";
        cout << endl;
    }
return 0;
}
```

Output:



12. Write a program to find solution of Fractional Knapsack instance.

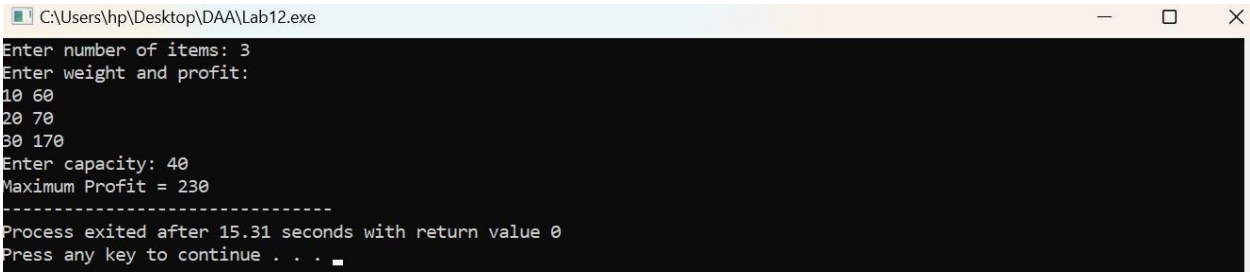
```
#include <iostream>
#include <algorithm>
using namespace std;
struct Item {
    int weight, profit;
    double ratio;
};
bool compare(Item a, Item b) {
    return a.ratio > b.ratio;
}
int main() {
    int n, capacity;
    double profit = 0;

    cout << "Enter number of items: ";
    cin >> n;
   Item a[n];
cout << "Enter weight and profit:\n";
    for (int i = 0; i < n; i++) {
        cin >> a[i].weight >> a[i].profit;
        a[i].ratio = (double)a[i].profit / a[i].weight;
    }
  cout << "Enter capacity: ";
    cin >> capacity;

    sort(a, a + n, compare);
```

```
    for (int i = 0; i < n && capacity > 0; i++) {
        if (a[i].weight <= capacity) {
            capacity -= a[i].weight;
            profit += a[i].profit;
        } else {
            profit += a[i].ratio * capacity;
            capacity = 0;
        }
    }
cout << "Maximum Profit = " << profit;
return 0;
}
```

Output:

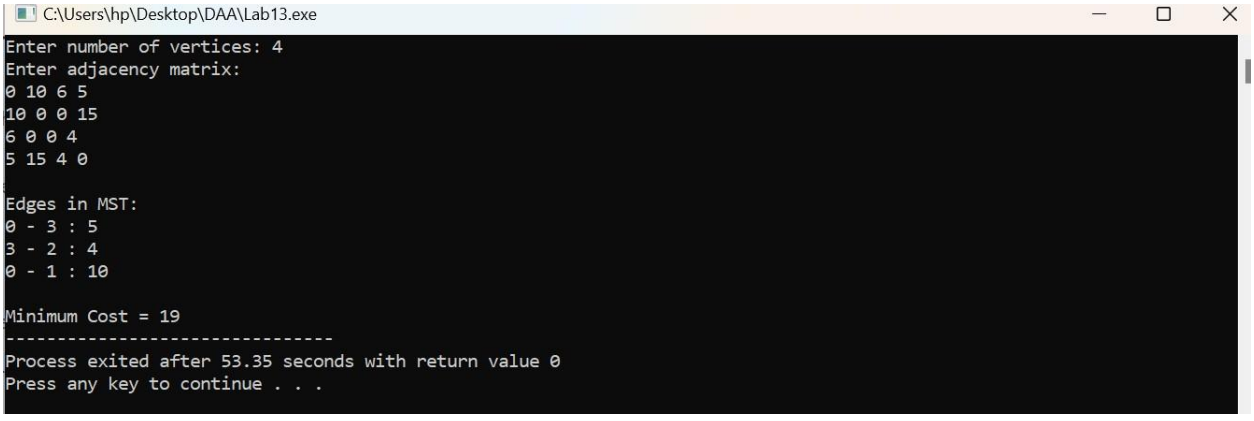


13. Write a program to find Minimum Spanning Tree using Prim's algorithm.

```
#include  <iostream>
using namespace std;
#define INF 9999
int main() {
   int n;
   cout << "Enter number of vertices: ";
   cin >> n;
int g[n][n], key[n], parent[n], used[n];
 cout << "Enter adjacency matrix:\n";
   for (int i = 0; i < n; i++)
      for (int j = 0; j < n; j++) {
         cin >> g[i][j];
         if (g[i][j] == 0)
            g[i][j] = INF;
      }
  for (int i = 0; i < n; i++) {
      key[i] = INF;
      parent[i] = -1;
      used[i] = 0;
   }
 key[0] = 0;
   int cost = 0;
 cout << "\nEdges in MST:\n";
for (int c = 0; c < n; c++) {
      int u = -1;
  for (int i = 0; i < n; i++)
```

```
            if (!used[i] && (u == -1 || key[i] < key[u]))
                u = i;
used[u] = 1;
 if (parent[u] != -1) {
            cout << parent[u] << " - " << u
                 << " : " << g[parent[u]][u] << endl;
            cost += g[parent[u]][u];
        }
  for (int v = 0; v < n; v++)
            if (!used[v] && g[u][v] < key[v]) {
                key[v] = g[u][v];
                parent[v] = u;
            }
    }
cout << "\nMinimum Cost = " << cost;
  return 0;
}
```

Output:

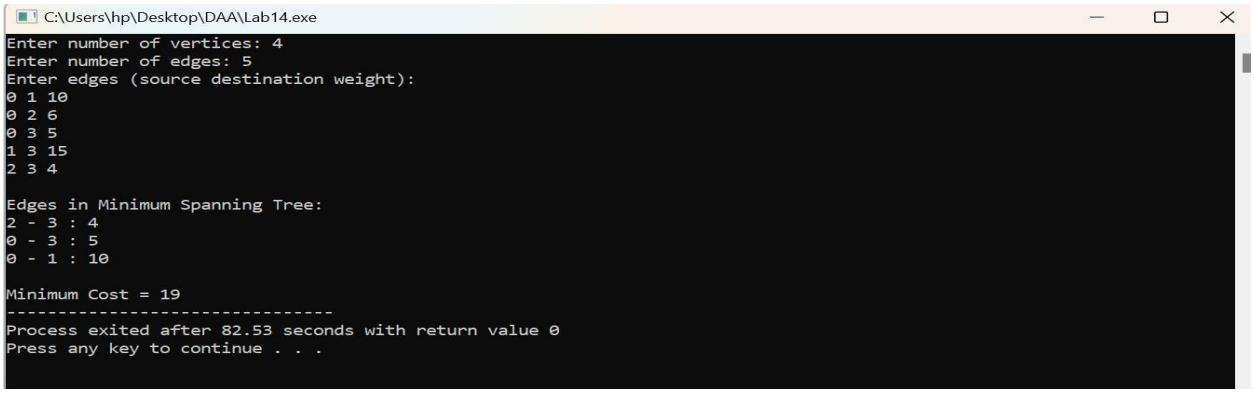


14. Write a program to find Minimum Spanning tree using Kruskal's algorithm.

```
#include <iostream>
#include <algorithm>
using namespace std;
struct Edge {
    int u, v, weight;
};
bool compare(Edge a, Edge b) {
    return a.weight < b.weight;
}int parent[100];
int find(int x) {
    if (parent[x] == x)
        return x;
    return parent[x] = find(parent[x]);
}void unionSet(int a, int b) {
    a = find(a);
    b = find(b);
 if (a != b)
        parent[b] = a;
}int main() {
    int n, e;
cout << "Enter number of vertices: ";
    cin >> n;
  cout << "Enter number of edges: ";
    cin >> e;
  Edge edges[e];
```

```
cout << "Enter edges (source destination weight):\n";
    for (int i = 0; i < e; i++)
        cin >> edges[i].u >> edges[i].v >> edges[i].weight;
  sort(edges, edges + e, compare);
for (int i = 0; i < n; i++)
        parent[i] = i;
int totalCost = 0, count = 0;
cout << "\nEdges in Minimum Spanning Tree:\n";
 for (int i = 0; i < e && count < n - 1; i++) {
         int u = edges[i].u;
        int v = edges[i].v;
if (find(u) != find(v)) {
            cout << u << " - " << v << " : "
                << edges[i].weight << endl;
  totalCost += edges[i].weight;
            unionSet(u, v);
            count++;
        }}cout << "\nMinimum Cost = " << totalCost;
return 0;}
```

Output:

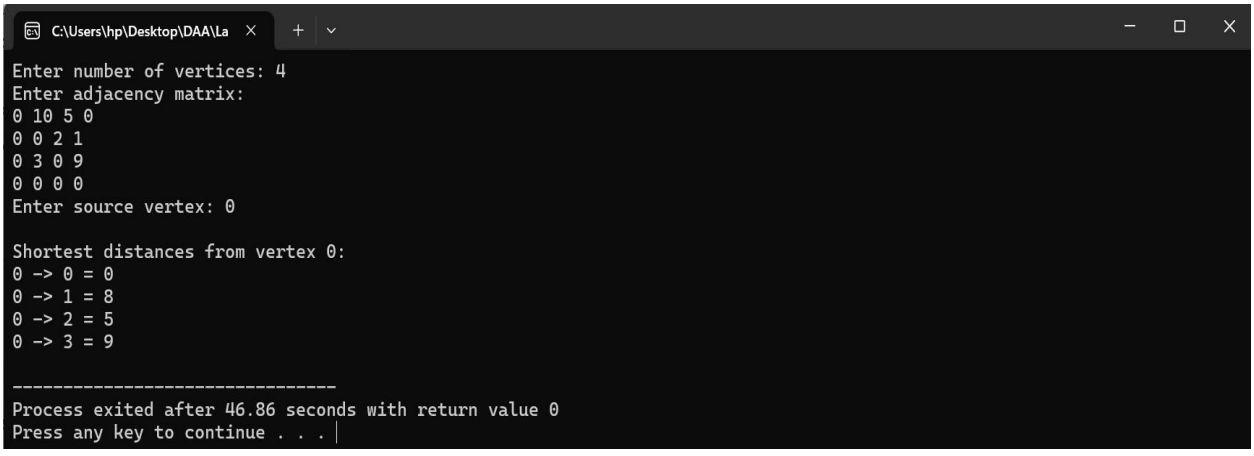


15. Write a program to find Single Source Shortest Path using Dijkstra'salgorithm

```
#include  <iostream>
using namespace std;
#define INF 9999
int main() {
   int n, source;
cout << "Enter number of vertices: ";
   cin >> n;
  int graph[n][n];
cout << "Enter adjacency matrix:\n";
   for (int i = 0; i < n; i++)
       for (int j = 0; j < n; j++) {
           cin >> graph[i][j];
  if (graph[i][j] == 0 && i != j)
               graph[i][j] = INF;
       }
cout << "Enter source vertex: ";
   cin >> source;
int distance[n], visited[n];
for (int i = 0; i < n; i++) {
       distance[i] = graph[source][i];
       visited[i] = 0;
   } distance[source] = 0;
for (int count = 0; count < n - 1; count++) {
       int u = -1, minDistance = INF;
for (int i = 0; i < n; i++)
```

```
  if (!visited[i] && distance[i] < minDistance) {
                minDistance = distance[i];
                u = i;
            }  if (u == -1)
            break;
visited[u] = 1;
 for (int v = 0; v < n; v++)
            if (!visited[v] && graph[u][v] != INF &&
                distance[u] + graph[u][v] < distance[v])
                distance[v] = distance[u] + graph[u][v];
    } cout << "\nShortest distances from vertex " << source << ":\n";
  for (int i = 0; i < n; i++) {
        if (distance[i] == INF)
            cout << source << " -> " << i << " = Not Reachable\n";
        else
            cout << source << " -> " << i << " = " << distance[i] << endl;
    }return 0;
}
```

Output:



16. Write a program to find solution of Knapsack Instance (0/1).

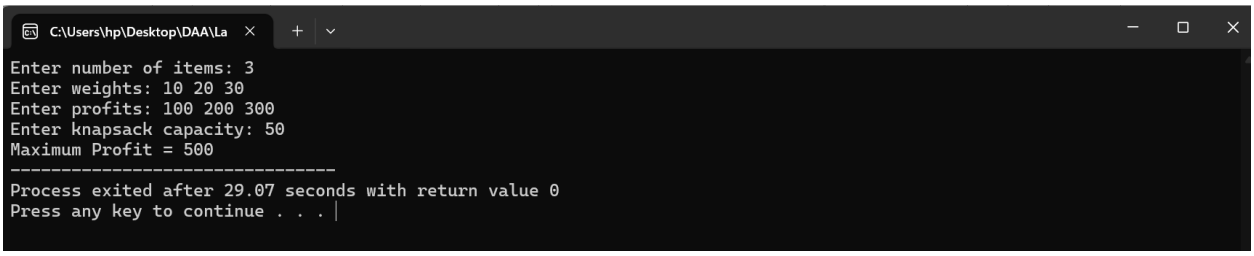
```
#include  <iostream>
using namespace std;
int main() {
    int n, capacity;
cout << "Enter number of items: ";
    cin >> n;
 int weight[n], profit[n];
  cout << "Enter weights: ";
  for (int i = 0; i < n; i++)
        cin >> weight[i];
 cout << "Enter profits: ";
    for (int i = 0; i < n; i++)
        cin >> profit[i];
cout << "Enter knapsack capacity: ";
    cin >> capacity;
    int dp[n + 1][capacity + 1];
   for (int i = 0; i <= n; i++) {
        for (int w = 0; w <= capacity; w++) {
  if (i == 0 || w == 0)
                dp[i][w] = 0;
  else if (weight[i - 1] <= w)
                dp[i][w] = max(profit[i - 1] +
                               dp[i - 1][w - weight[i - 1]],
                               dp[i - 1][w]);
 else
                dp[i][w] = dp[i - 1][w];
```

```
}
   }

   cout << "Maximum Profit = " << dp[n][capacity];

   return 0;
}
```

Output:



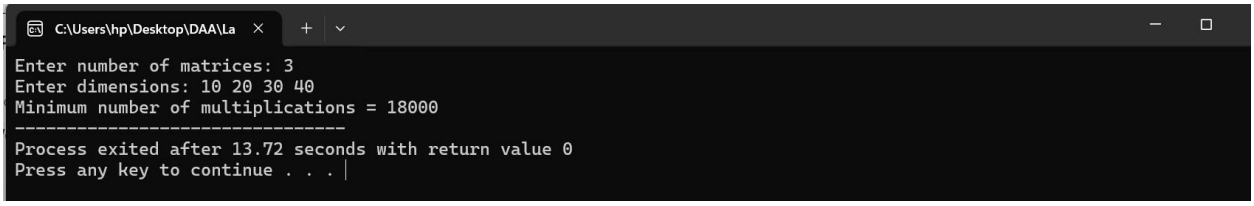
17. Write a program to find solution of Matrix Chain Multiplication.

```
#include  <iostream>
using namespace std;
int main() {
    int n;
  cout << "Enter number of matrices: ";
    cin >> n;
 int p[n + 1];
 cout << "Enter dimensions: ";
    for (int i = 0; i <= n; i++)
        cin >> p[i];
int dp[n + 1][n + 1];
for (int i = 1; i <= n; i++)
        dp[i][i] = 0;
for (int len = 2; len <= n; len++) {
        for (int i = 1; i <= n - len + 1; i++) {
            int j = i + len - 1;
            dp[i][j] = 999999999;
for (int k = i; k < j; k++) {
                int cost = dp[i][k] + dp[k + 1][j]
                        + p[i - 1] * p[k] * p[j];
if (cost < dp[i][j])
                    dp[i][j] = cost;
            }
        }
    }
```

```
    cout << "Minimum number of multiplications = "
         << dp[1][n];

    return 0;
}
```

Output:

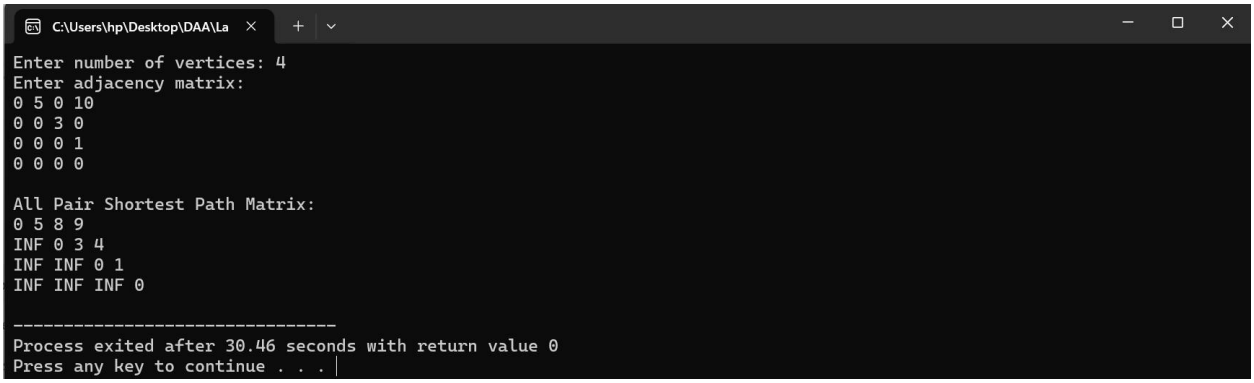


18. Write a program to find shortest path using All Pair Shortest Path algorithm.

```
#include  <iostream>
using namespace std;
#define INF 9999
int main() {
   int n;
cout << "Enter number of vertices: ";
   cin >> n;
int dist[n][n];
 cout << "Enter adjacency matrix:\n";
   for (int i = 0; i < n; i++)
       for (int j = 0; j < n; j++) {
          cin >> dist[i][j];
  if (dist[i][j] == 0 && i != j)
              dist[i][j] = INF;
       }
for (int k = 0; k < n; k++)
       for (int i = 0; i < n; i++)
          for (int j = 0; j < n; j++)
             if (dist[i][k] != INF && dist[k][j] != INF &&
                dist[i][k] + dist[k][j] < dist[i][j])
                dist[i][j] = dist[i][k] + dist[k][j];
cout << "\nAll Pair Shortest Path Matrix:\n";
for (int i = 0; i < n; i++) {
       for (int j = 0; j < n; j++) {
          if (dist[i][j] == INF)
```

```
            cout << "INF ";
        else
            cout << dist[i][j] << " ";
    }
    cout << endl;
  }
return 0;
}
```

Output:



19. Write a program to Traverse Graph – Depth First Search, Breadth First Search.

```
#include <iostream>
using namespace std;
int graph[10][10], visited[10], n;
void DFS(int v) {
    cout << v << " ";
    visited[v] = 1;
for (int i = 0; i < n; i++)
        if (graph[v][i] == 1 && !visited[i])
            DFS(i);
}
void BFS(int start) {
    int queue[10], front = 0, rear = 0;

    for (int i = 0; i < n; i++)
        visited[i] = 0;
 queue[rear++] = start;
     visited[start] = 1;
  while (front < rear) {
        int v = queue[front++];
        cout << v << " ";
   for (int i = 0; i < n; i++) {
            if (graph[v][i] == 1 && !visited[i]) {
                queue[rear++] = i;
                visited[i] = 1;
            }} }}
```

```
int main() {
    int start;
 cout << "Enter number of vertices: ";
    cin >> n;
 cout << "Enter adjacency matrix:\n";
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            cin >> graph[i][j];
 cout << "Enter starting vertex: ";
    cin >> start;
for (int i = 0; i < n; i++)
        visited[i] = 0;
 cout << "\nDFS: ";
    DFS(start);
cout << "\nBFS: ";
    BFS(start);
return 0;
}
```

Output:



20. Write a program to find all solutions for N-Queen problem using backtracking.

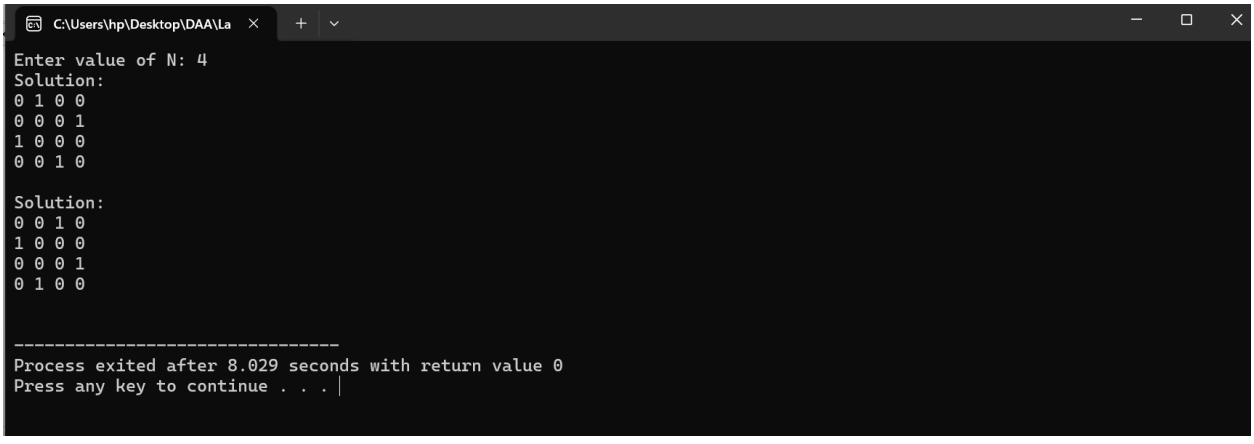
```
#include  <iostream>
using namespace std;
int board[20][20], n;
bool isSafe(int row, int col) {
    for (int i = 0; i < row; i++)
        if (board[i][col])
            return false;
  for (int i = row - 1, j = col - 1; i >= 0 && j >= 0; i--, j--)
        if (board[i][j])
            return false;
for (int i = row - 1, j = col + 1; i >= 0 && j < n; i--, j++)
        if (board[i][j])
            return false;
 return true;
}
void solve(int row) {
    if (row == n) {
        cout << "Solution:\n";
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++)
                cout << board[i][j] << " ";
            cout << endl;
        }
        cout << endl;
        return;
```

```
  }
for (int col = 0; col < n; col++) {
        if (isSafe(row, col)) {
            board[row][col] = 1;
            solve(row + 1);
            board[row][col] = 0;
        }
    }
}
int main() {
    cout << "Enter value of N: ";
    cin >> n;

    solve(0);

    return 0;
}
```

Output:



21. Write a program for Graph Colouring using backtracking.

```
#include <iostream>
using namespace std;
int graph[20][20], color[20], n, m;
bool isSafe(int v, int c) {
    for (int i = 0; i < n; i++)
        if (graph[v][i] && color[i] == c)
            return false;
    return true;
}
bool graphColoring(int v) {
    if (v == n)
        return true;
for (int c = 1; c <= m; c++) {
        if (isSafe(v, c)) {
            color[v] = c;
if (graphColoring(v + 1))
                return true;

            color[v] = 0;
        } }
return false;
}
int main() {
    cout << "Enter number of vertices: ";
    cin >> n;
```

```
    cout << "Enter adjacency matrix:\n";
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            cin >> graph[i][j];

    cout << "Enter number of colors: ";
    cin >> m;

    if (graphColoring(0)) {
        cout << "\nGraph Coloring:\n";
        for (int i = 0; i < n; i++)
            cout << "Vertex " << i << " = Color " << color[i] << endl;
    }
    else
        cout << "Graph cannot be colored using " << m << " colors.";

    return 0;
}
```

Output:

